

Paper	ENCN342 – Fluid Mechanics and Hydraulics
Topic	Steady Open Channel Flow
Module	3 – The Hydraulic Jump

Summary

This module introduces the hydraulic jump and explains why it may occur. The momentum equation is used to analyse the hydraulic jump.

Prerequisites

Module 2.

Objectives

At the conclusion of this module you should:

1. Understand how two channel controls can be in conflict.
2. Understand the concept of a hydraulic jump and when it might form.
3. Be able to analyse a hydraulic jump using the M - y diagram and the hydraulic jump equation.

References

Hunt 1995 pp 4.16-4.20

Key Points

1. Conflicting controls

A control

- fixes a depth-discharge relationship in its vicinity
- has subcritical flow upstream and supercritical flow downstream
- extends its influence upstream and downstream.

If two controls are present in a channel their influences are generally in **conflict** in the channel reach between them.

This conflict may be resolved by the presence of a hydraulic jump.

2. The hydraulic jump

A **hydraulic jump** corresponds to a rapid transition from supercritical flow to subcritical flow that is accompanied by a significant energy loss.

The momentum equation must be used to analyse the hydraulic jump, due to the energy loss involved. The ratio of the depths upstream and downstream of the jump are given by the hydraulic jump equation

$$\frac{y_2}{y_1} = \frac{1}{2} \left(\sqrt{1 + 8Fr_1^2} - 1 \right) \quad (1)$$

or

$$\frac{y_1}{y_2} = \frac{1}{2} \left(\sqrt{1 + 8Fr_2^2} - 1 \right) \quad (2)$$

depending on whether the upstream or downstream conditions are known. The depths y_1 and y_2 are known as **conjugate depths**.

3. Momentum function and the M-y diagram

The **momentum function**, M , is defined by

$$M = \frac{q^2}{gy} + \frac{1}{2}y^2 \quad (3)$$

It is analogous to the specific energy, E . For flow regions in which no other significant forces, except the hydrostatic thrust, are present we have

$$M_1 = M_2 \quad (4)$$

For flows in which an upstream force per unit width, p_f , is present equation (4) becomes

$$M_1 - M_2 = \frac{p_f}{\rho g} \quad (5)$$

An M - y diagram, or dimensionless M' - y' diagram, can be used to analyse open channel flow problems with the momentum equation. M' is defined as

$$M' = \frac{M}{y_c^2} \quad (6)$$

So

$$M' = \frac{1}{y'} + \frac{1}{2} y'^2 \quad (7)$$

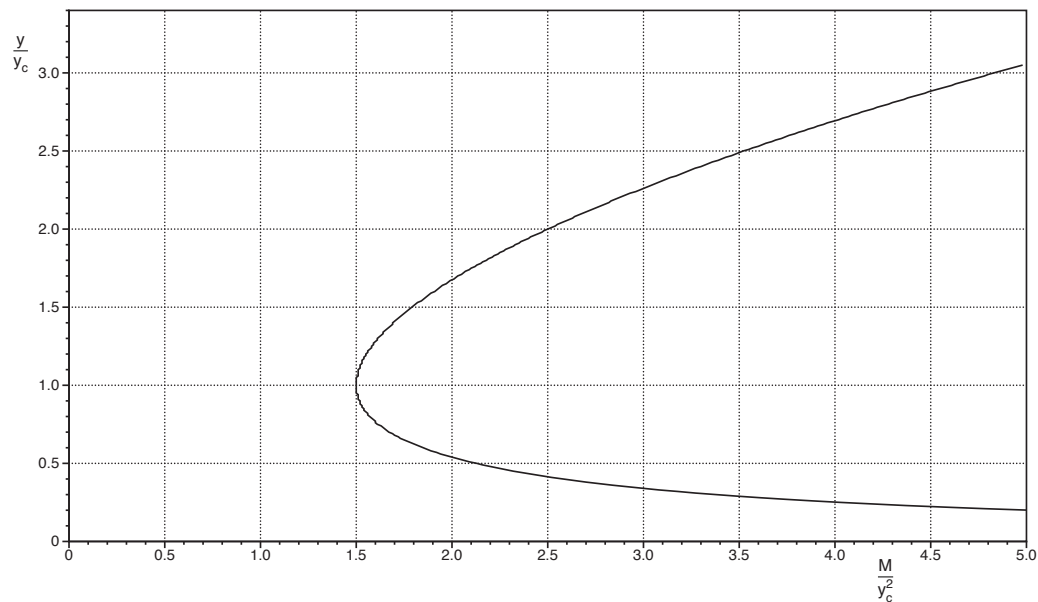


Figure 3.1 The dimensionless M' - y' curve

Worked Examples

Question 1

A rectangular channel carrying a discharge per unit width of q encounters a smooth step up in channel bed elevation of Δz . If the flow upstream of the step is y_1 and the depth on the step is y_2 , calculate the force per unit width that the flow exerts on the step, in terms of y_1 , y_2 and Δz (ie eliminate q from your expression).

Solution

The specific energy changes with the change in bed elevation. Therefore

$$E_1 = y_1 + \frac{q^2}{2gy_1^2} = E_2 + \Delta z = y_2 + \frac{q^2}{2gy_2^2} + \Delta z$$

This equation can be rearranged to find q in terms of the depths and bed step.

$$q^2 = \frac{2g(y_2 + \Delta z - y_1)}{\left(\frac{1}{y_1^2} - \frac{1}{y_2^2}\right)} = \frac{2gy_1^2 y_2^2 (y_2 + \Delta z - y_1)}{(y_2^2 - y_1^2)}$$

Now the force per unit width on the step can be calculated from the difference in momentum function upstream and downstream of the step.

$$\begin{aligned} F_{step} &= (M_1 - M_2)\rho g = \rho g \left(\frac{q^2}{gy_1} + \frac{1}{2}y_1^2 - \frac{q^2}{gy_2} - \frac{1}{2}y_2^2 \right) \\ &= \rho g \left(\frac{q^2}{g} \left(\frac{1}{y_1} - \frac{1}{y_2} \right) + \frac{1}{2}y_1^2 - \frac{1}{2}y_2^2 \right) \\ &= \rho g \left(\frac{2y_1^2 y_2^2 (y_2 + \Delta z - y_1)}{(y_2^2 - y_1^2)} \left(\frac{1}{y_1} - \frac{1}{y_2} \right) + \frac{1}{2}y_1^2 - \frac{1}{2}y_2^2 \right) \\ &= \rho g \left(\frac{2y_1^2 y_2^2 (y_2 + \Delta z - y_1)}{(y_2 - y_1)(y_2 + y_1)} \left(\frac{y_2 - y_1}{y_1 y_2} \right) + \frac{1}{2}y_1^2 - \frac{1}{2}y_2^2 \right) \\ &= \rho g \left(\frac{2y_1 y_2 (y_2 + \Delta z - y_1)}{(y_2 + y_1)} + \frac{1}{2}y_1^2 - \frac{1}{2}y_2^2 \right) \end{aligned}$$

The algebraic manipulations could be taken further, but this equation can be used to calculate the force purely in terms of the depths, and the step.

Question 2

A sluice gate is mounted in a channel of width $5m$. Upstream of the gate the flow depth is $3m$. The gate opening is $1m$, and the contraction coefficient is 0.61 .

Downstream of the sluice gate a change in channel bed slope causes the flow to make a transition from supercritical flow to subcritical flow through a hydraulic jump.

- (a) Calculate the flow discharge in the channel.
- (b) Calculate the depth downstream of the jump.
- (c) Calculate the force on the sluice gate.
- (d) Calculate the energy loss (in metres) in the jump.

Solution

(a) For the flow under the sluice gate there are negligible energy losses so the specific energy should be the same upstream and downstream of the gate. If y_1 is the upstream depth and y_2 the downstream depth then

$$E = y_1 + \frac{q^2}{2gy_1^2} = y_2 + \frac{q^2}{2gy_2^2}$$

So solving this equation for q we have

$$q^2 = \frac{2g(y_1 - y_2)}{\frac{1}{y_2^2} - \frac{1}{y_1^2}}$$

Now the upstream depth is $3m$, while the downstream depth is the product of the gate opening and the contraction factor. Thus

$$y_2 = 1m \times 0.61 = 0.61m$$

Therefore the discharge is

$$q^2 = \frac{2g(3 - 0.61)}{\frac{1}{0.61^2} - \frac{1}{3^2}} = 18.2m^4 / s^2$$

Thus

$$q = 4.27m^2 / s$$

and

$$Q = qB = 4.27 \times 5 = 21.4m^3 / s$$

(b) We can calculate the depth downstream of the jump, y_2 , using the hydraulic jump equation. We know the conditions upstream so we use the equation involving the upstream Froude number which we will designate Fr_2 .

$$Fr_2 = \frac{u_2}{\sqrt{gy_2}} = \frac{q}{\sqrt{gy_2^3}} = \frac{4.27}{\sqrt{9.81 \times 0.61^3}} = 2.86$$

Now

$$\frac{y_3}{y_2} = \frac{1}{2} \left(\sqrt{1 + 8Fr_2^2} - 1 \right) = 3.58$$

So the downstream depth is

$$y_3 = 3.58 \times y_2 = 2.18m$$

(c) As there are no energy losses through the gate the specific energy is the same on either side of the gate. However, because there is a net force on the gate due to the upstream flow, the momentum function is not the same. By calculating the difference between the momentum function upstream and downstream of the gate the force of the gate on the fluid can be calculated, and hence the force of the fluid on the gate.

$$M_1 = \frac{q^2}{gy_1} + \frac{1}{2}y_1^2 = \frac{18.2}{3 \times 9.81} + 4.5 = 5.12$$

$$M_2 = \frac{q^2}{gy_2} + \frac{1}{2}y_2^2 = \frac{18.2}{0.61 \times 9.81} + 0.39 = 3.23$$

Thus the net force on the fluid (and hence the gate, except in the opposite direction of course) is

$$F_{gate} = (M_1 - M_2) \rho g B = 1.89 \times 9810 \times 5 = 92.7kN$$

(d) Finally we wish to calculate the energy loss through the jump. The momentum function has the same value either side of the jump, but the specific energy does not. The change in specific energy provides the energy loss through the jump.

$$E_2 = y_2 + \frac{q^2}{2gy_2^2} = 0.61 + \frac{18.2}{2 \times 9.81 \times 0.61^2} = 3.10$$

$$E_3 = y_3 + \frac{q^2}{2gy_3^2} = 2.18 + \frac{18.2}{2 \times 9.81 \times 2.18^2} = 2.38$$

Therefore the energy loss through the jump is

$$E_{loss} = E_2 - E_3 = 3.10 - 2.38 = 0.72m$$